B: CONFIGURATION CHOICES

Digital twin configuration: separate facets, complementary responsibilities

A literature-informed synthesis; descriptors may coexist and do not imply uncertainty capability.

System composition

Unit and combination

Single, aggregate, composite; network, federation, ecosystem.

Uncertainty responsibility

Preserve entity identities and shared dependencies when combining components or stocks.

Software and semantics

Software organisation

Layered, modular, service-oriented.

Knowledge representation

Knowledge graphs / semantic links.

Uncertainty responsibility

Retain variable meaning, units, provenance and interface contracts.

Computational model

Model basis

Physics-based, data-driven, hybrid.

Scale (a separate attribute)

Single-scale or multiscale scope.

Uncertainty responsibility

Specify input distributions, dependencies and transformations.

Coupling and authority

Data exchange

One-way or bidirectional.

Decision / control authority

Human-mediated or automated.

Uncertainty responsibility

Define what is transferred and who may approve a revision.

Agency and simulation

Agent-based simulation

Models interacting entities.

Multi-agent orchestration

Coordinates software services.

Agentic AI (proposed here)

Goal-directed tool orchestration; not a substitute for UQ or testing.

Temporal organisation

Update trigger and cadence

Periodic, event-triggered or continuous; single / multiple rates.

Timing constraints (separate)

Latency, freshness and validity.

Uncertainty responsibility

Invalidate stale dependent outputs.

No universal compatibility or performance is inferred from selecting these descriptors.

Grouping, scale/cadence attributes and S1–S5 responsibilities are the review synthesis.